

COSMOLOGICAL OBSERVABLES

Tomislav Prokopec, ITP Utrecht University

References:

Miao, Prokopec, Woodard, "Deducing Cosmological Observables from the S-matrix," Phys. Rev. **D** 96 (2017) 10, 104029 [1708.06239 [gr-qc]]

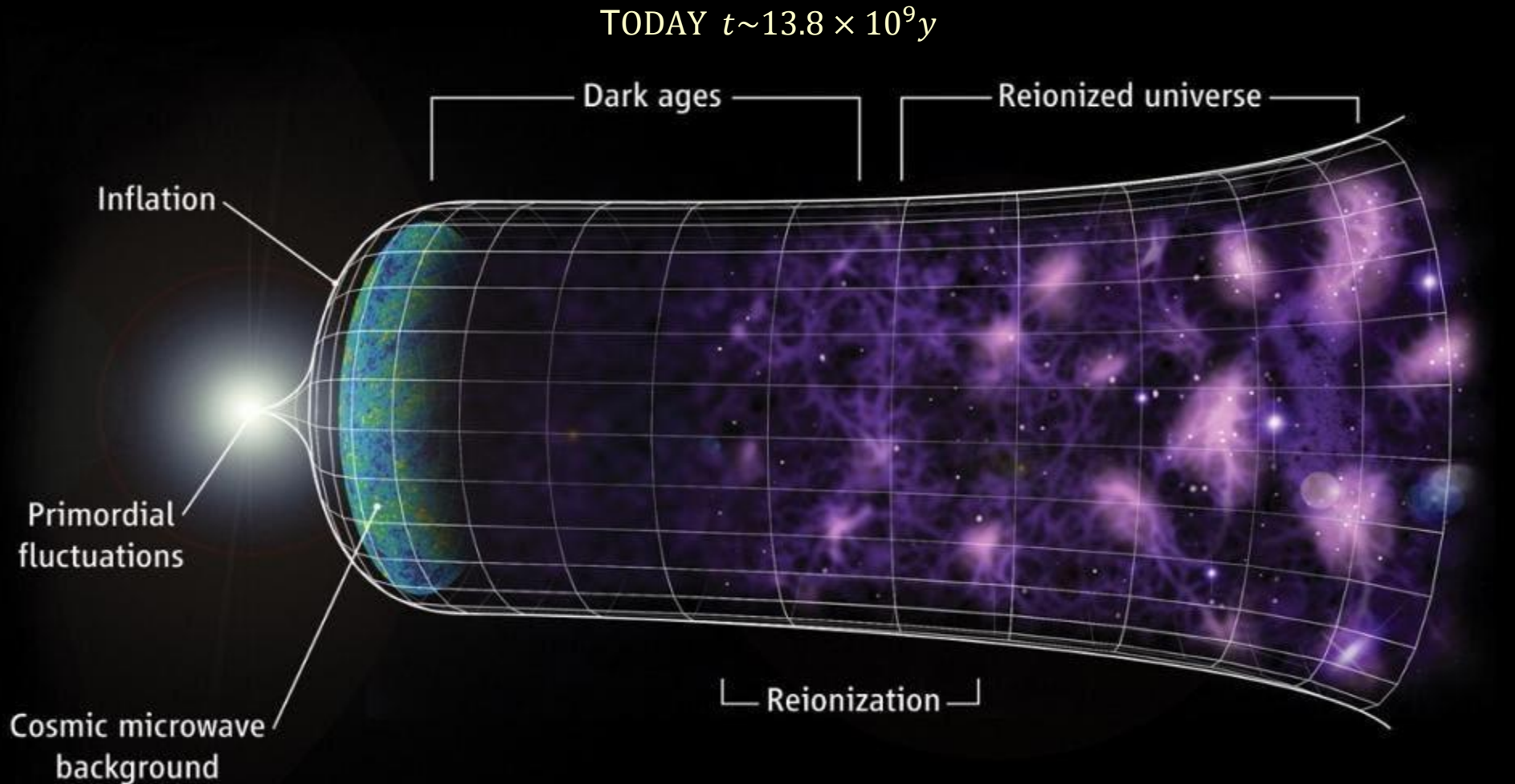
Glavan, Miao, Prokopec, Woodard, e-Print: 2602.07908 [hep-th], 2609.10365 [gr-qc]

Zagreb, 23 Sep 2026

HISTORY OF THE UNIVERSE

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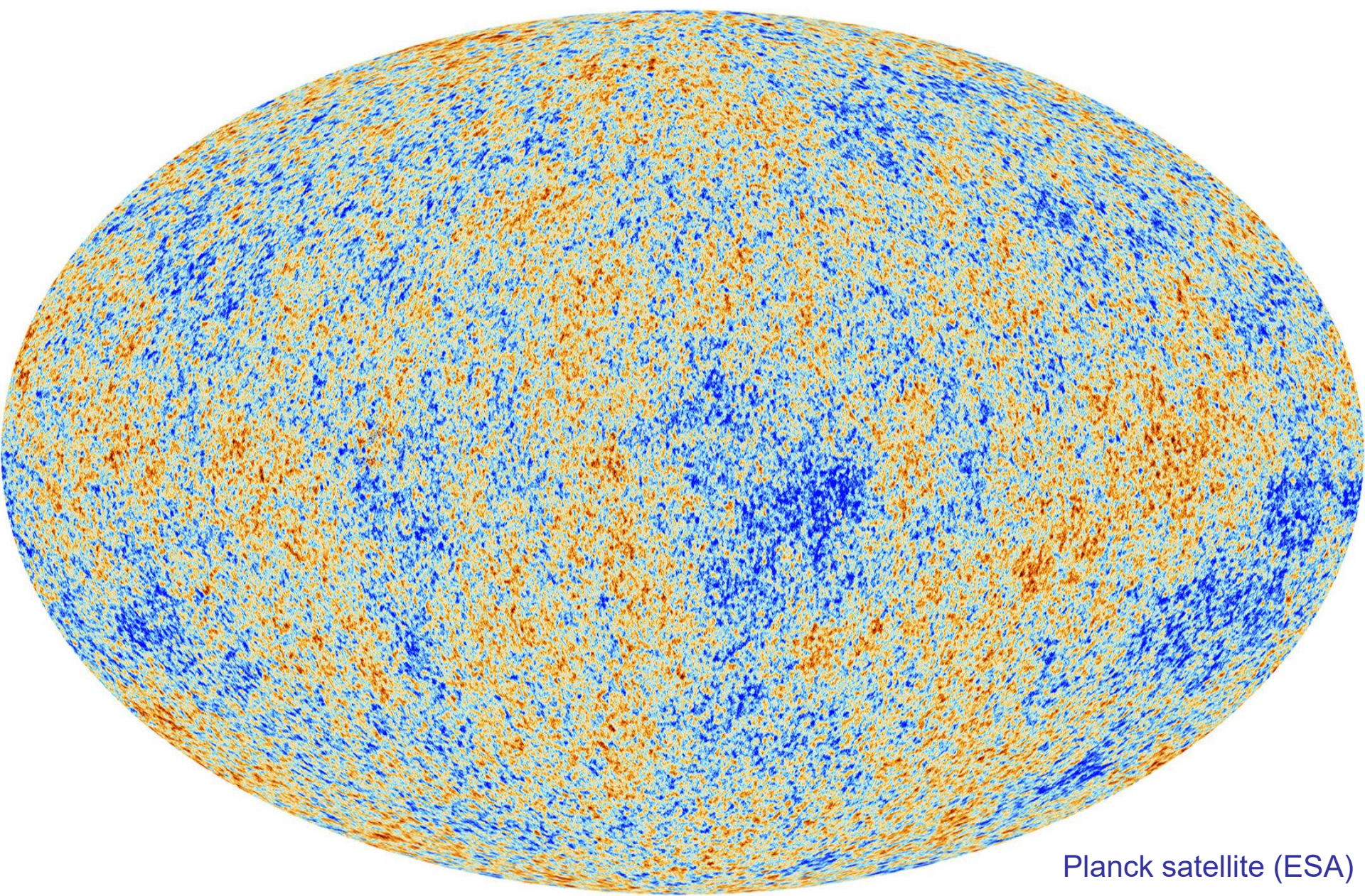
INFLATION → RADIATION ERA → MATTER ERA → DARK ENERGY



INFLATION: $t \sim 10^{-34} s$ MATTER ERA: $t \sim 40000 y$ PHOTON DECOUPLING (CMB): $t \sim 3.8 \times 10^5 y$

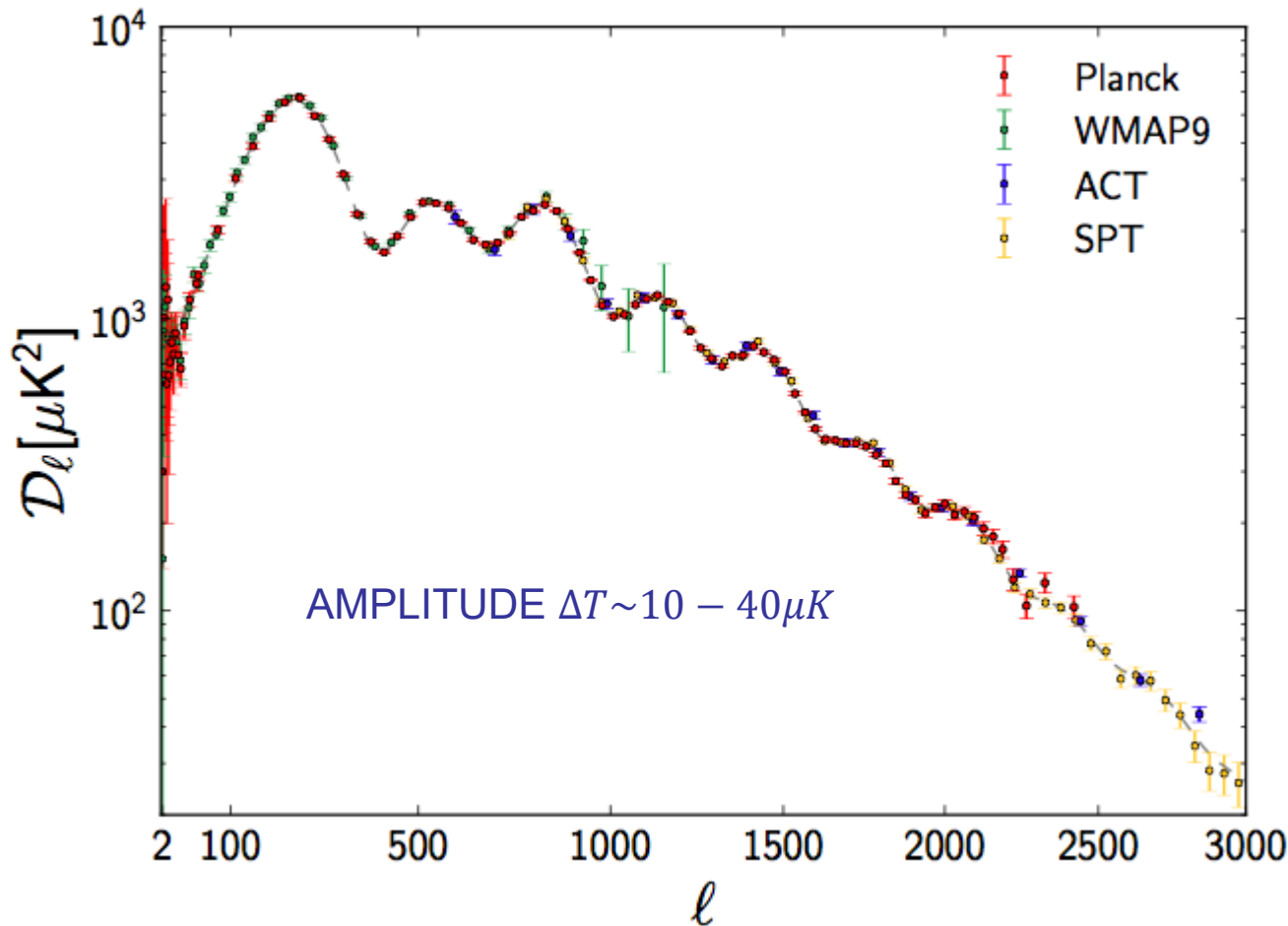
WHAT DO WE OBSERVE?

COSMIC MICROWAVE RADIATION (CMB)



Planck satellite (ESA)

CMB SPECTRUM



SPECTRUM -- STATISTICAL
AVERAGE:

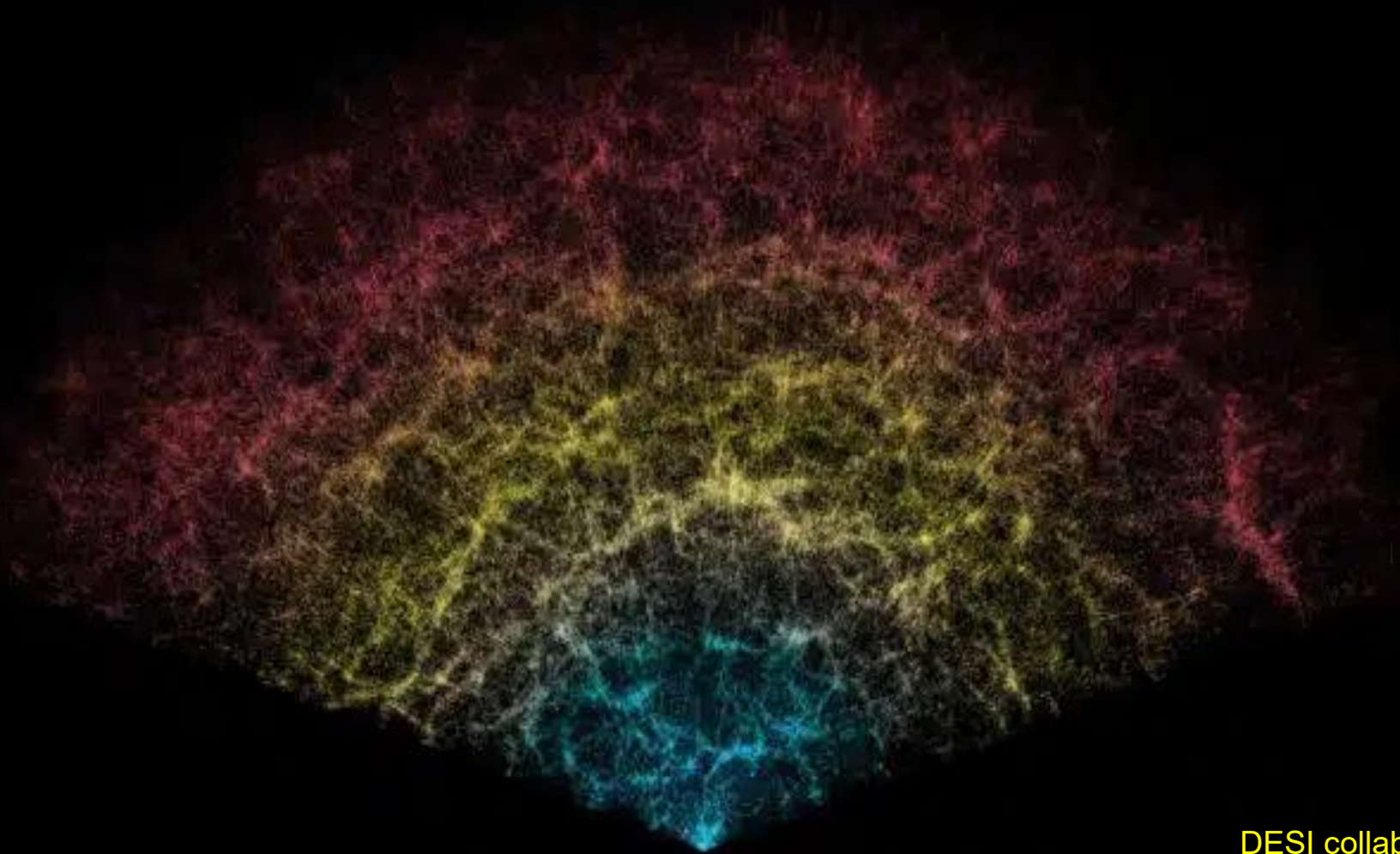
$$\langle \Delta \hat{T}(\hat{n}) \Delta \hat{T}(\hat{n}') \rangle$$

l = angular scale
 $l \sim 180 \equiv 1^\circ$

- THE SPECTRUM SHOWS ACOUSTIC PEAKS: THE ECHOES FROM THE PRIMORDIAL UNIVERSE
- THE SPECTRUM IS CHARACTERISED BY TWO NUMBERS: THE AMPLITUDE AND SLOPE
- FROM THE SPECTRUM ONE CAN READ OFF THE EXPANSION HISTORY: $H(t)$ [$\Omega_{\text{dm}}, \Omega_{\text{b}}, \Omega_{\text{v}}, \Omega_{\gamma}$]

LARGE SCALE STRUCTURE (LSS)

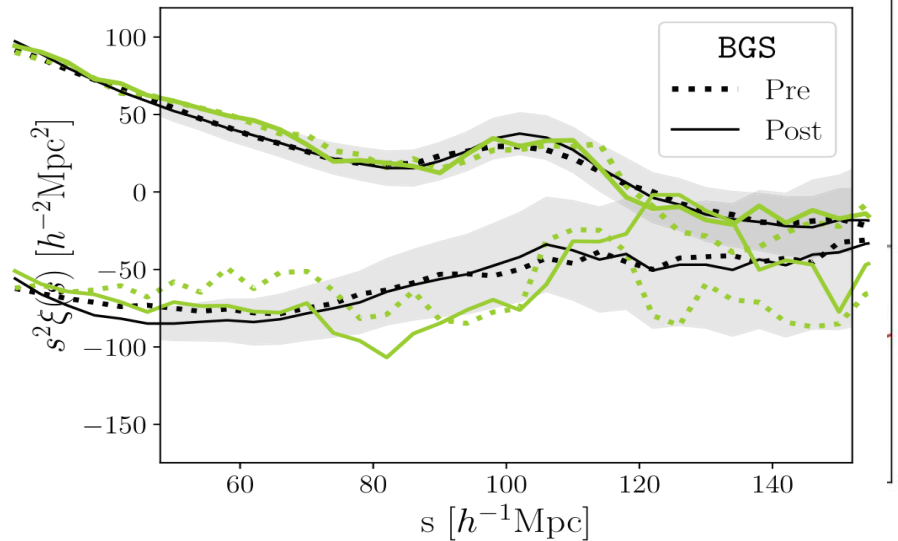
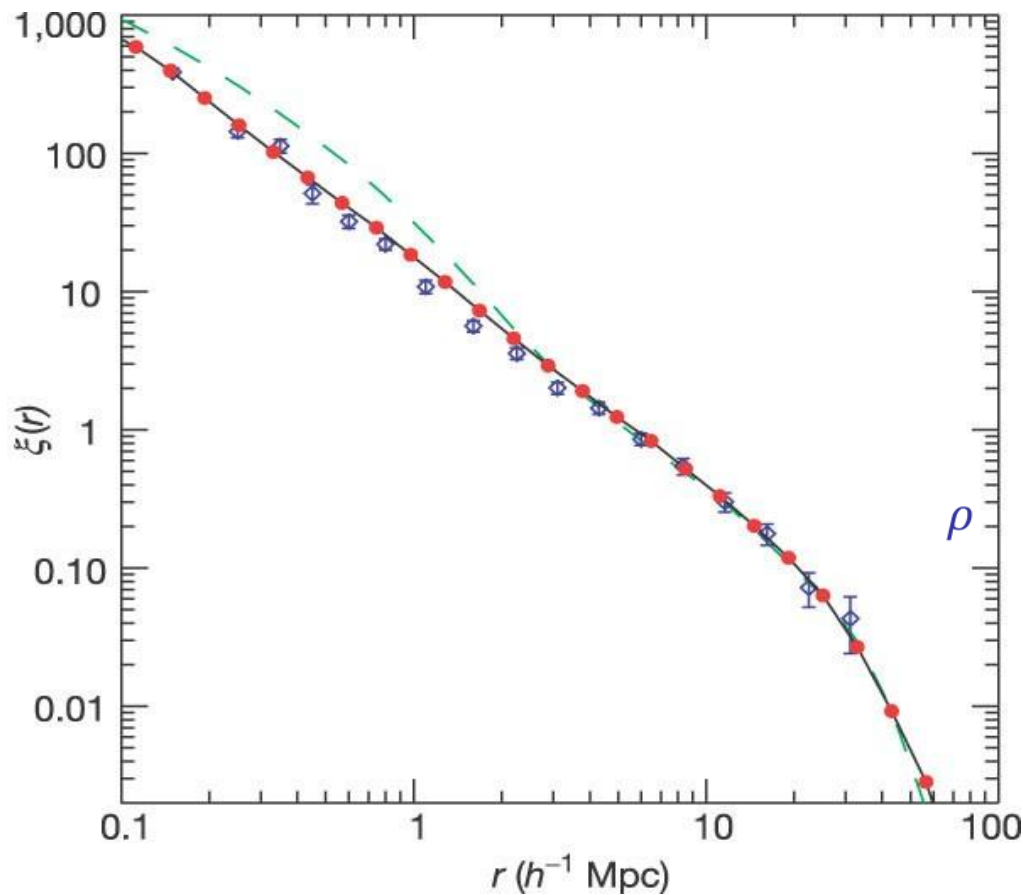
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DESI collaboration

PRESENTLY: EUCLID (satellite) and LSST (Vera Rubin, Željko Ivezić) ARE TAKING DATA.

LSS SPECTRUM



BAO PEAK: DESI collaboration (2025)

GALAXY CORRELATION FUNCTION $\xi(r)$ (2005)

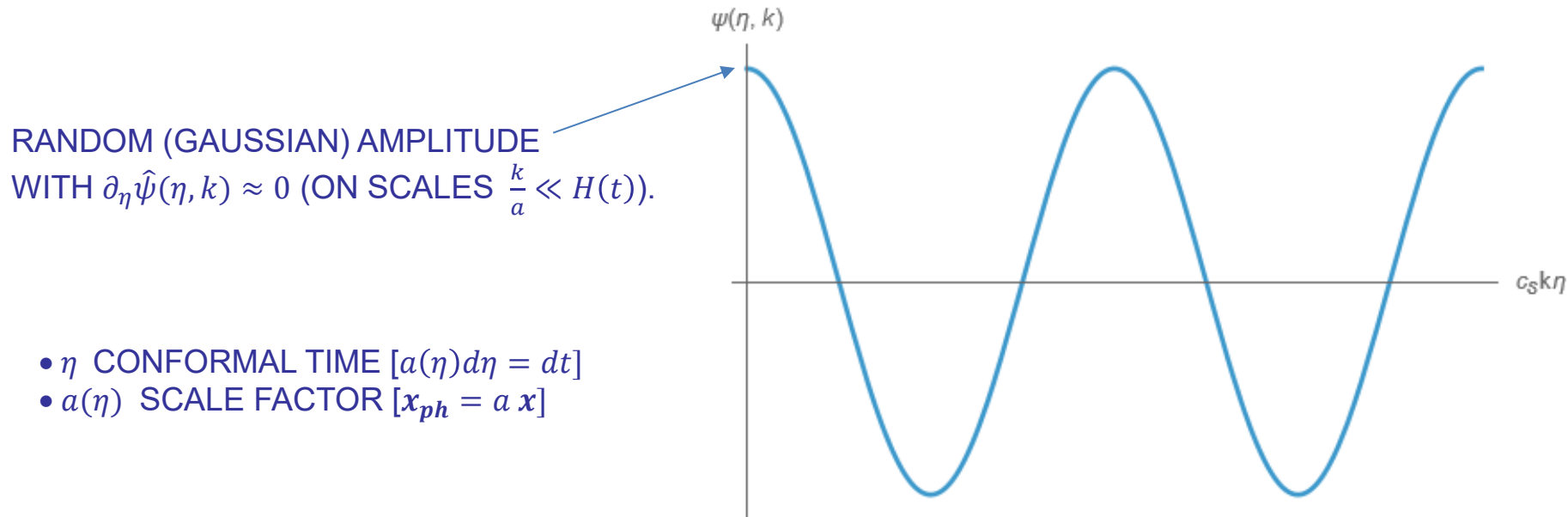
THESE SPECTRA CONTAIN INFORMATION ABOUT SEED GRAVITATIONAL FIELDS ψ THAT CREATED THEM.

$$\left\langle \frac{\delta \hat{\rho}(t, \mathbf{x})}{\bar{\rho}(t)} \frac{\delta \hat{\rho}(t, \mathbf{x}')}{\bar{\rho}(t)} \right\rangle \propto \langle \nabla^2 \hat{\psi}(t, \mathbf{x}) \nabla'^2 \hat{\psi}(t, \mathbf{x}') \rangle \quad \text{(POISSON EQUATION)}$$

SEED GRAVITATIONAL FIELDS

° 8°

- AT THE BEGINNING OF MATTER ERA, GRAVITATIONAL POTENTIAL EXHIBITS SAKHAROV OSCILLATIONS:



- GRAVITATIONAL POTENTIAL $\hat{\psi}$
 - (a) DRAWN FROM A RANDOM GAUSSIAN ENSEMBLE
 - (b) APPROXIMATELY FLAT SPECTRUM
 - (c) $\partial_\eta \hat{\psi} \approx 0$

SUGGESTS QUANTUM ORIGIN
⇒ COSMIC INFLATION
[period of a nearly exponentially
expanding universe]

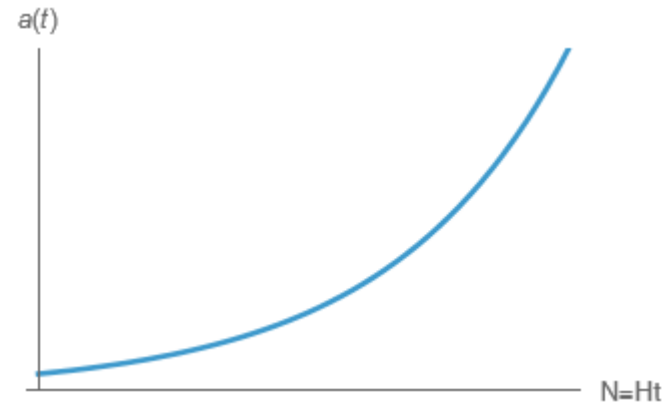
COSMOLOGICAL PERTURBATIONS

COSMOLOGICAL PERTURBATIONS

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- COSMOLOGICAL INFLATION: HYPOTHETICAL PERIOD OF ALMOST EXPONENTIAL EXPANSION OF THE UNIVERSE:

- scale factor: $a(t) \sim e^{Ht}$, $H \sim \text{const.}$



- VACUUM FLUCTUATIONS OF MATTER FIELDS GET STRETCHED & AMPLIFIED
⇒ GENERATES GRAVITATIONAL POTENTIAL $\hat{\psi}$ WHICH IS:
 - (a) RANDOM GAUSSIAN FIELD
 - (b) THE SPECTRUM IS NEARLY SCALE INVARIANT
 - (c) ON LARGE SCALES $\partial_t \hat{\psi} \sim 0$

EXACTLY AS
OBSERVED!

GRAVITATIONAL POTENTIALS

- GRAVITY IS DEFINED BY THE METRIC TENSOR $g_{\mu\nu}(x)$
(USED TO MEASURE DISTANCES ON SPACETIME)
- QUANTUM METRIC PERTURBATIONS $\delta\hat{g}_{\mu\nu}(x) = \text{diag}(2\hat{\phi}, 2\hat{\psi}, 2\hat{\psi}, 2\hat{\psi})$
CAN BE DEFINED IN TERMS OF 2 GRAVITATIONAL POTENTIALS
 - $\phi(t, \mathbf{x})$: responsible for time dilations [$d\tau = \sqrt{1 + 2\phi} dt$]
→ used to synchronize clocks on satellites and Earth for GPS;
 - $\psi(t, \mathbf{x})$: responsible for distance contractions [$dl = a \sqrt{1 - 2\psi} dr$]
→ since on Earth $\psi > 0$, physical distances are contracted.

INFLATIONARY AMPLIFICATION

- INFLATION AMPLIFIES QUANTUM VACUUM FLUCTUATIONS OF MATTER FIELDS. WE SEE IT IN THE GRAVITATIONAL FIELD $\hat{\psi}$:
- $(\partial_\eta^2 + (2 - \epsilon_2)aH\partial_\eta + k^2)\hat{\psi}(\eta, k) = 0$ [ϵ_2 = acceleration of $H(t)$]
HUBBLE DAMPING: $(2 - \epsilon_2)aH$ RESPONSIBLE FOR THE AMPLIFICATION
- WE UNDERSTAND HOW TO CONNECT THE SPECTRUM OF $\hat{\psi}(\eta, k)$ TO THE LATE TIME OBSERVABLES:
→ CMB TEMPERATURE PERTURBATIONS,
→ GALAXY CORRELATION FUNCTION OF LSS).

BUT: WE ONLY KNOW HOW TO DO THAT IF WE NEGLECT INTERACTIONS OF $\hat{\psi}$ WITH MATTER AND GRAVITY!

- INCLUDING THE INTERACTIONS WOULD NOT ONLY ALLOW US TO BETTER MODEL THE SPECTRUM OF $\hat{\psi}$, BUT IT WOULD ALSO GIVE US IMPORTANT INFORMATION ABOUT THE STATE AND CONTENT OF MATTER & GRAVITY IN THE PRIMORDIAL UNIVERSE.

INTERACTIONS WITH MATTER & GRAVITY

- INTERACTIONS ARE CONTROLLED BY THE LOOP COUNTING PARAMETER OF QUANTUM GRAVITY:

- $\kappa^2 = \frac{16 \pi G \hbar}{c^5}$. THIS GIVES FOR INFLATION $\kappa^2 H^2 \sim 10^{-10}$ (1/10 billion, a tiny number)
- WHY BOTHER!?

- THE COUNTING IS MISLEADING FOR TWO REASONS:

- appearance of secular effects in inflation: $\sim \ln^N(a) \sim (Ht)^N$ [N=1,2,..] (Weinberg)
- the counting can be misleading and in radiation & matter era can be much larger
Ota, Sasaki, Wang (2023); Liu, Prokopec (2024); Fröb, Glavan, Sawicki (2025)
- in matter era (in fact in general), when $\frac{\delta \rho}{\bar{\rho}} \sim 1$, the interactions become strong (PBH).

PROBLEM: WE DO NOT KNOW HOW TO COMPUTE THESE INTERACTIONS
IN A GAUGE INDEPENDENT WAY. BUT, WHAT IS GAUGE DEPENDENCE IN GRAVITY?

- IN GRAVITY GAUGE DEPENDENCE IS JUST THE DEPENDENCE ON THE OBSERVER.
 $\hat{\psi}$ -- and thus its spectrum -- ARE OBSERVER DEPENDENT.

GAUGE SYMMETRY IN ELECTROMAGNETISM

- WE DESCRIBE ELECTROMAGNETISM BY GAUGE FIELD $A^\mu = (A^0, \mathbf{A}) \equiv (\phi, \mathbf{A})$. THIS DESCRIPTION IS HOWEVER REDUNDANT: ONE COMPONENT IS GAUGE!
- ANOTHER WAY OF STATING IT IS: PHYSICAL FIELDS ARE INDEPENDENT ON THE TRANSFORMATION:

$$A^\mu \rightarrow A^\mu + \partial^\mu \Lambda(t, \mathbf{x}), \quad \text{WITH } \Lambda(t, \mathbf{x}) \text{ ARBITRARY FUNCTION.}$$

E.G.: ELECTRIC FIELD: $\mathbf{E} = -\nabla A^0 - \partial_t \mathbf{A} \Rightarrow \mathbf{E}^L = -\nabla^L A^0 - \partial_t \mathbf{A}^L$ ONLY ONE OF THE TWO COMPONENTS $A^0, \mathbf{A}^L$ IS PHYSICAL $\Rightarrow$ ONE COMPONENT IS REDUNDANT.

ONE CAN FIX A GAUGE BY, FOR EXAMPLE, FIXING $A^0 = 0$, OR $\mathbf{A}^L = 0$.

IN BOTH CASES ONE ENDS UP WITH THE SAME ELECTRIC FIELD $\mathbf{E}$, WHICH IS THE PHYSICAL QUANTITY. IN ELECTROMAGNETISM, THIS GAUGE FIXING IS FIXING INTERNAL COMPONENTS OF THE ELECTROMAGNETIC POTENTIAL, AND IT IS THUS WELL UNDERSTOOD.

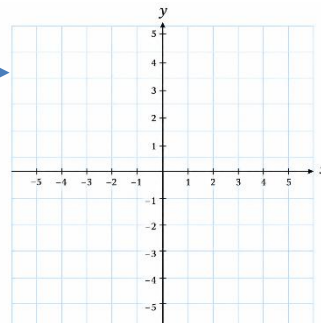
IN GRAVITY GAUGE DEPENDENCE IS THE DEPENDENCE ON THE DEFINITION OF COORDINATE SYSTEM, EQUIVALENTLY OF OBSERVER. DIFFERENT OBSERVERS CAN PERCEIVE DIFFERENT GRAVITATIONAL POTENTIALS $\hat{\psi}$. WHICH ONE IS THEN THE ONE WE PERCEIVE IN LATE TIME OBSERVATIONS?

COORDINATE SYSTEMS $\rightarrow$

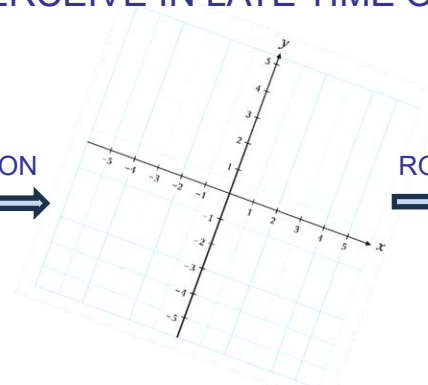
COORDINATE SHIFT:

$$x^\mu \rightarrow \tilde{x}^\mu = x^\mu + \xi^\mu(x)$$

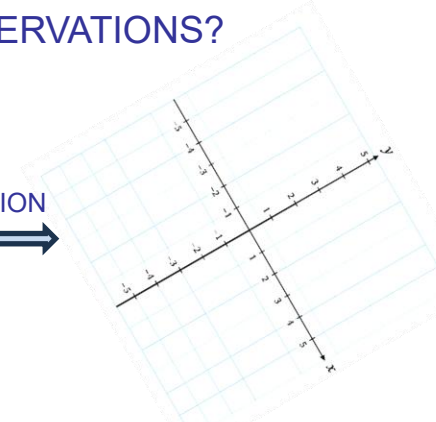
$\Rightarrow$ GAUGE TRANSFORMATION



ROTATION



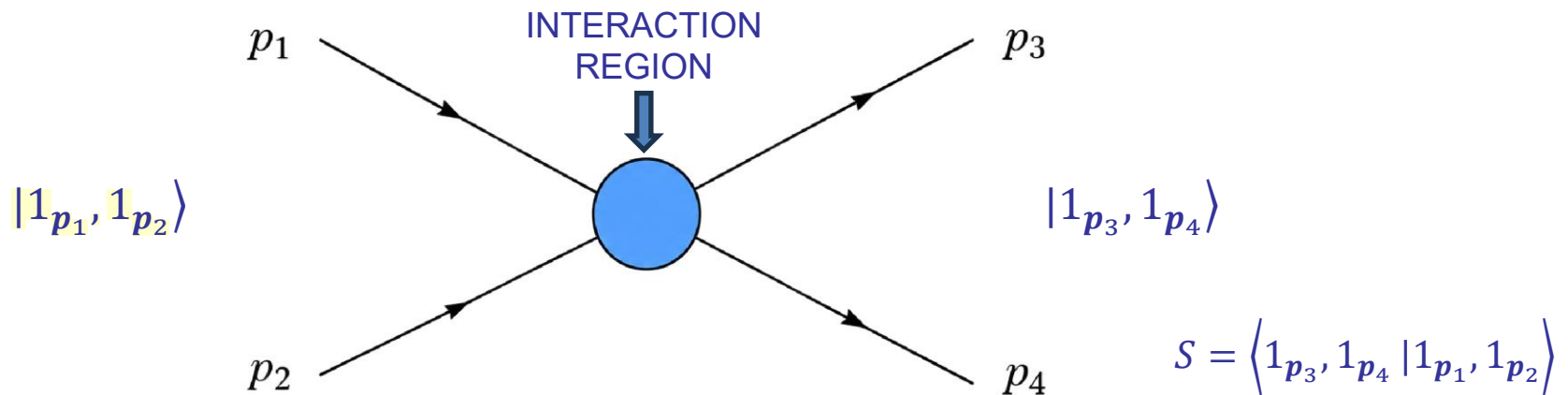
ROTATION



THE PROBLEM OF GAUGE DEPENDENCE EXISTS ALSO IN THE STANDARD MODEL (QFT), AND IT HAS BEEN SOLVED BY THE SO-CALLED SCATTERING (S-) MATRIX FORMALISM.

S-MATRIX

- IMAGINE YOU PREPARE A STATE (at $t \rightarrow -\infty$) WITH TWO PARTICLES $|1_{p_1}, 1_{p_2}\rangle$. YOU ARE INTERESTED IN THE FOLLOWING QUESTION: WHAT IS THE PROBABILITY AMPLITUDE (OR CROSS SECTION) THAT THE FINAL STATE (at $t \rightarrow +\infty$) $|1_{p_3}, 1_{p_4}\rangle$ WILL ALSO CONTAIN TWO PARTICLES?



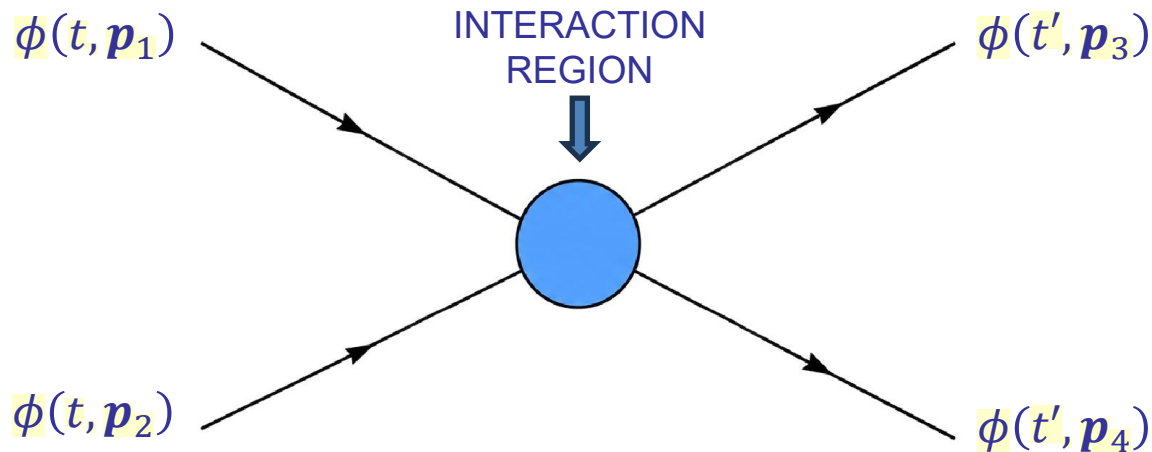
THE SCATTERING MATRIX (**S-MATRIX**) FORMALISM TELLS US THAT THE ANSWER YOU GET AFTER THE CALCULATION WILL BE INDEPENDENT ON THE GAUGE YOU CHOOSE!

Conditions: asymptotic states exist and interactions are localized, i.e. asymptotic states can be taken to be noninteracting. Photons and gravitons mediate long interactions, so for electromagnetism and gravity interactions are not localized. However, one can solve this problem by suitably redefining asymptotic states (by performing a constant re-normalization on them).

In COSMOLOGY (and in general curved spaces): neither asymptotic states exist, nor they can be taken to be noninteracting.

SCATTERING IN COSMOLOGY

- AN INITIAL STATE $(\phi(t, \mathbf{p}_1), \phi(t, \mathbf{p}_2))$ IS PREPARED AT FINITE TIME t . THE QUESTION IS: WHAT IS THE PROBABILITY AMPLITUDE/CROSS SECTION FOR THE FINAL STATE AT $t' > t$ TO BE $(\phi^*(t', \mathbf{p}_3), \phi^*(t', \mathbf{p}_4))$.



THE STATES $\phi(t, \mathbf{p}_i)$ ($i = 1, \dots, 4$) SATISFY THE CLASSICAL (TREE-LEVEL) EQUATION OF MOTION. E.g. FOR A SCALAR FIELD $\phi(t, \mathbf{x})$ THIS IS THE KLEIN-GORDON (WAVE) EQUATION:

$$(\square_i - m^2)\phi(t, \mathbf{x}_i) = 0 \Rightarrow (\partial_\eta^2 + 2aH\partial_\eta + \|\mathbf{p}_i\|^2 + a^2m^2)\phi(\eta, \mathbf{p}_i) = 0 \quad (i = 1, 2)$$

SUMMARY: ASYMPTOTIC STATES OF THE S-MATRIX ARE REPLACED BY THE CLASSICAL EQUATIONS OF MOTION (ON-SHELL RELATIONS).

(FOR GRAVITONS, THE WAVE EQUATION IS MORE COMPLICATED.)

- In our paper in 2017 [Miao, Prokopec, Woodard] we showed that this works in [flat Minkowski space](#).

MINKOWSKI SPACE CALCULATION

°17°

Miao, Prokopec, Woodard (2017)

INSPIRED BY WORKS OF BJERRUM-BOHR, DONOGHUE & HOLSTEIN IN THE EARLY 2000s. THEY COMPUTED THE 1-LOOP QUANTUM GRAVITON CORRECTION TO THE NEWTONIAN POTENTIAL:

$$\phi_N(r) = \underbrace{-\frac{Gm}{r}}_{\text{CLASSICAL POTENTIAL}} \left[1 + \underbrace{3\frac{Gm}{c^2 r}}_{\text{RELATIVISTIC CORRECTION}} + \underbrace{\frac{41}{10\pi} \frac{G\hbar}{c^3 r^2}}_{\text{1-LOOP CORRECTION (tiny)}} + \dots \right]$$

WE COMPUTED THE 1-LOOP CORRECTION TO THE MASSLESS SCALAR POTENTIAL:

- EFFECTIVE SCALAR FIELD EQUATION OF MOTION:

$$\partial^2 \phi(x) - \int d^4 x' M_{\text{ret}}^2(x; x') \phi(x') = q \delta^3(x)$$

$$M_{\text{ret}}^2(x; x') = -\frac{3G\hbar}{8\pi^2 c^3} \partial^8 \{ \Theta((t - t') - \|\mathbf{x} - \mathbf{x}'\|) [\log(\mu^2[(t - t')^2 - \|\mathbf{x} - \mathbf{x}'\|^2])] - 1 \}$$

- THE 1-LOOP QUANTUM GRAVITON CORRECTION TO THE SCALAR FIELD POTENTIAL HAS THE SAME STRUCTURE:

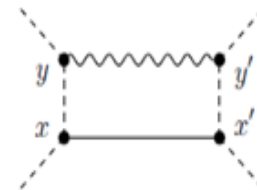
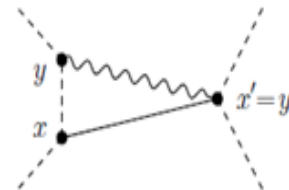
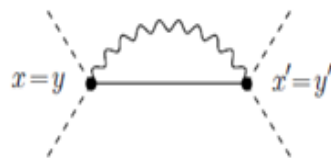
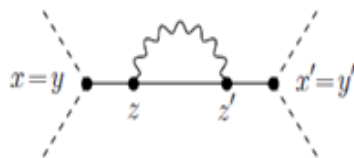
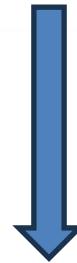
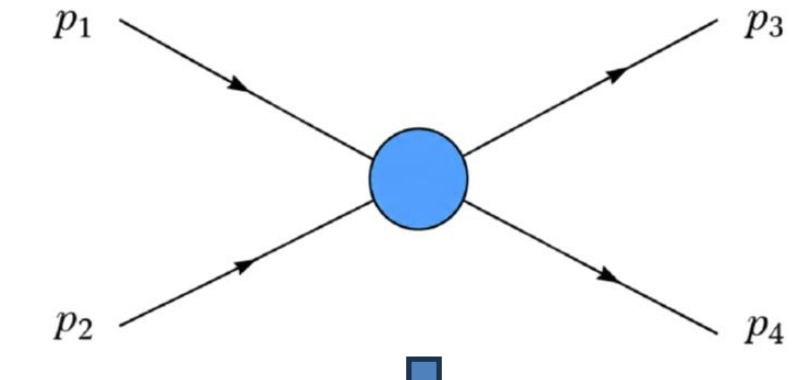
$$\phi(r) = -\frac{q}{4\pi r} \left[1 + \#_1 \frac{Gm}{c^2 r} + \#_2 \frac{G\hbar}{c^3 r^2} + \dots \right]$$

- WE HAVE USED THE MOST GENERAL COVARIANT GRAVITON PROPAGATOR AND SHOWED THAT THE RESULT DOES NOT DEPEND ON THE GAUGE PARAMETERS (α, β) , $\alpha, \beta \in \mathbb{R}$.

FEYNMAN DIAGRAMS IN MINKOWSKI

Miao, Prokopec, Woodard (2017)

CLASSES OF FEYNMAN DIAGRAMS AT THE 1-LOOP ORDER:



DE SITTER SPACE CALCULATION

°19°

Glavan, Miao, Prokopec, Woodard (2026)

WE HAVE RECENTLY REPEATED THE CALCULATION IN DE SITTER (INFLATION) AND SHOWED THAT THE RESULT **DOES NOT DEPEND** ON THE GAUGE PARAMETER $\alpha \in \mathbb{R}$.

$$\phi(r) = -\frac{q}{4\pi ar} \left[\underbrace{1 + ar \log\left(\frac{a}{1 + aHr}\right)}_{\text{CLASSICAL POTENTIAL}} + \underbrace{\#_1 \frac{G\hbar}{c^3 a^2 r^2}}_{\text{1-LOOP CONFORMAL CORRECTION}} + \underbrace{\#_2 G\hbar H^2 \log(aHr) + \dots}_{\text{DE SITTER CORRECTION}} \right]$$

WE COMPUTED THE 1-LOOP CORRECTION TO THE MASSLESS SCALAR POTENTIAL.

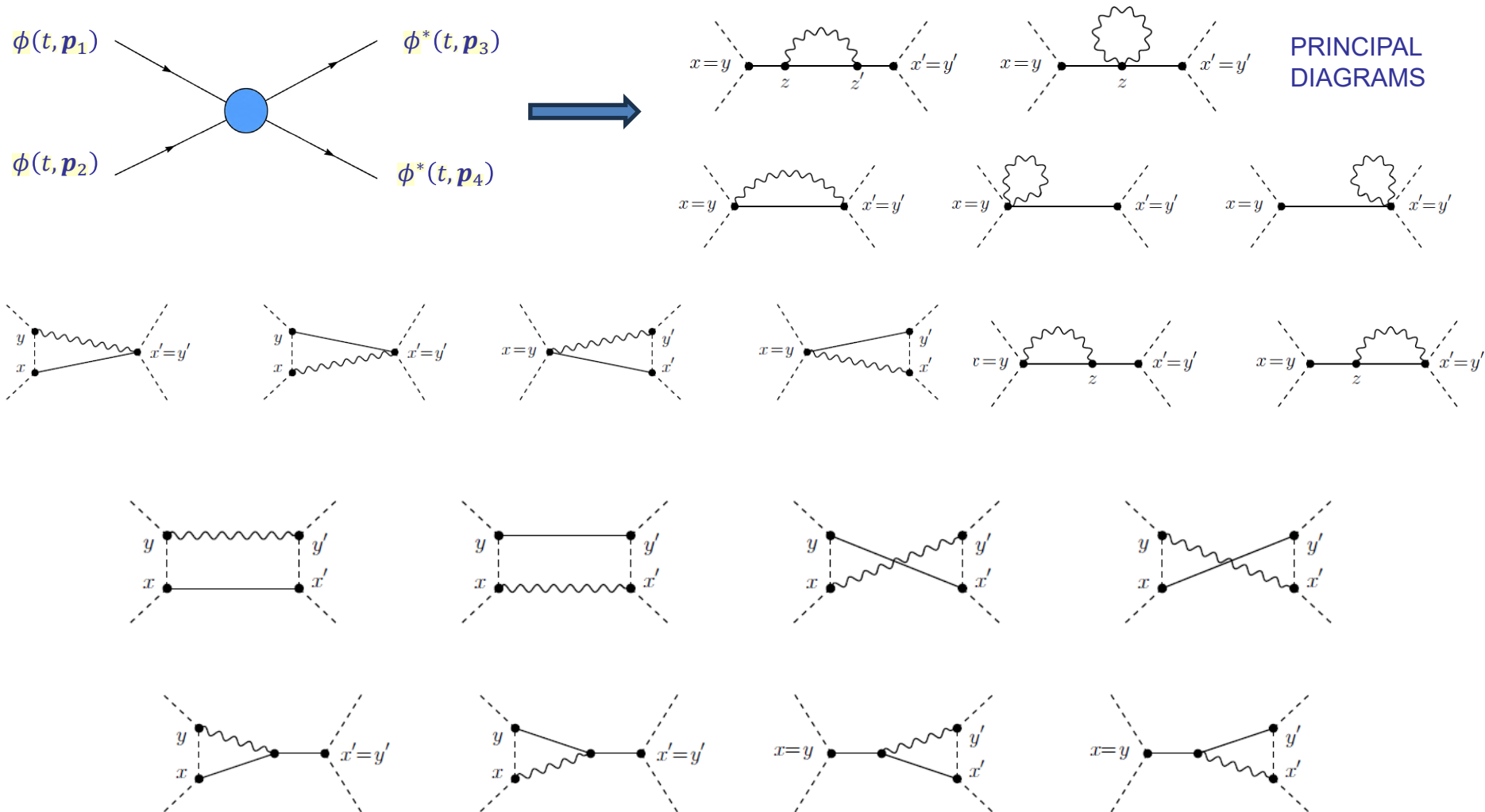
- EFFECTIVE SCALAR FIELD EQUATION OF MOTION:

$$\square \phi(x) - \int d^4 x' M_{\text{ret}}^2(x; x') \phi(x') = q \delta^3(x)$$

$M_{\text{ret}}^2(x; x')$ = de Sitter version of the scalar selfmass (complicated, contains $\sim \log(a)$)

FEYNMAN DIAGRAMS IN DE SITTER

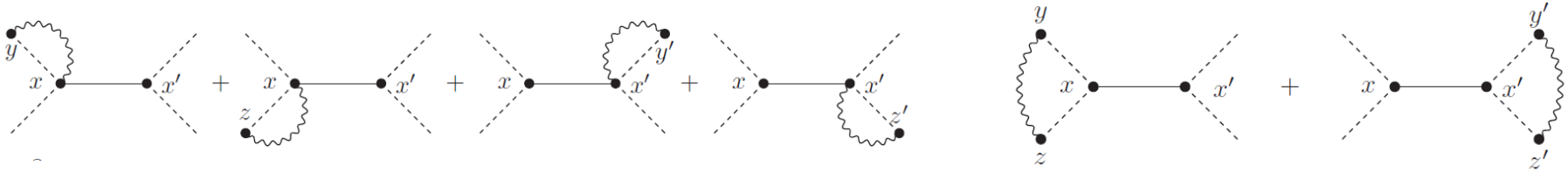
FEYNMAN DIAGRAMS AT THE 1-LOOP ORDER IN DE SITTER:



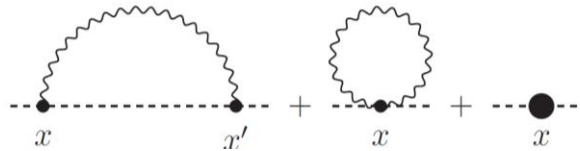
HOWEVER, THIS IS **NOT ENOUGH** TO MAKE THE ANSWER GAUGE INDEPENDENT!

FEYNMAN DIAGRAMS IN DE SITTER 2

IN CURVED SPACES (SUCH AS DE SITTER) NEW CLASSES OF DIAGRAMS ARE NEEDED (WHICH **DO NOT** CONTRIBUTE IN MINKOWSKI):

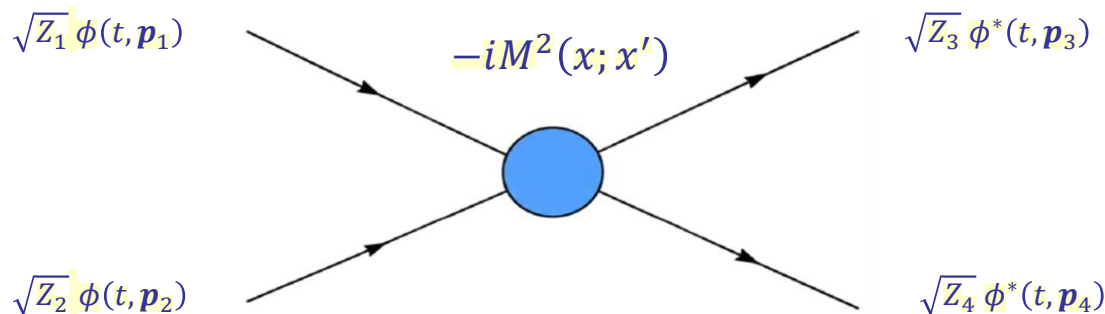


PLUS THE WAVE FUNCTION CORRECTION DIAGRAMS (WHICH IN MINKOWSKI CONTRIBUTE A CONSTANT):



THESE DIAGRAMS CORRECT THE EXTERNAL WAVE FUNCTIONS: $\phi(t, \mathbf{p}_i) \rightarrow \sqrt{Z_i} \phi(t, \mathbf{p}_i)$
AND IN ADDITION CONTRIBUTE A SECULAR TERM $\propto \ln(a)$ TO THE PRINCIPAL DIAGRAMS!

- A GENERALIZED FRAMEWORK FOR GETTING GAUGE INDEPENDENT ANSWERS IN CURVED SPACETIMES (SUCH AS COSMOLOGY):

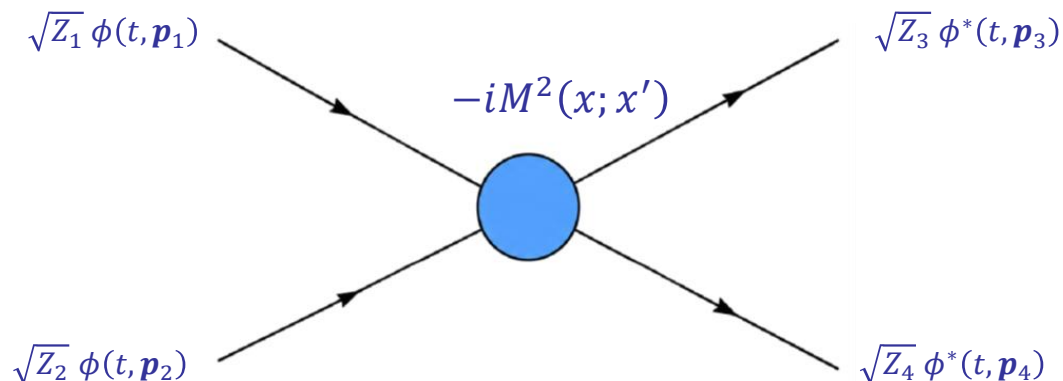


$-iM^2(x; x')$ RENORMALIZED
GAUGE INDEPENDENT
SELF-MASS

NEXT STEPS

- THE RENORMALIZED SELF-MASS $-iM^2(x; x')$ DOES NOT DEPEND ON ONE GAUGE PARAMETER α . CHECK THAT IT DOES NOT DEPEND ON β .
- COMPUTE THE QUANTUM 1-LOOP GRAVITATIONAL CORRECTIONS TO THE SCALAR 1-POINT FUNCTION $\phi(x)$ AND 2-POINT FUNCTION $i\Delta(x; x')$ (~INFLATON SPECTRUM).
CONJECTURE: $\delta P_{\Delta\phi}(t, k) \sim GH^2 \log^2(a) \approx GH^2 (Ht)^2 = GH^2 (Ht)^2 \sim 10^{-6}$. STILL TOO SMALL.
- EXTEND THE ANALYSIS TO (DYNAMICAL) GRAVITONS. EXPECTED ENHANCEMENT LARGER, $\sim GH^2 \log^N(a)$ ($N \geq 3$) $\Rightarrow$ POTENTIALLY OBSERVABLE
- REPEAT THE ANALYSIS IN A MORE REALISTIC INFLATIONARY MODEL, IN WHICH SCALAR INFLATON PERTURBATIONS COUPLE TO GRAVITATIONAL PERTURBATIONS.
- EXTEND THE ANALYSIS TO THE LATER EPOCHS: RADIATION ERA & MATTER ERA

WHEN COMPLETED, THIS PROGRAM WILL GIVE US AN **EFFECTIVE, GAUGE INDEPENDENT THEORY OF COSMOLOGICAL PERTURBATIONS** FOR THE EARLY UNIVERSE, AKIN TO THOSE THAT EXIST FOR ELECTROMAGNETISM AND OTHER MATTER FIELDS.



SUCH A THEORY IS OUR **BEST HOPE** TO PROBE THE PHYSICS OF THE EARLY UNIVERSE, AND TO **TEST PERTURBATIVE QUANTUM GRAVITY**.

GRAVITON SELF-ENERGY

- AT 1 LOOP ON DE SITTER WE HAVE

$$-i[\mu\nu\Sigma^{\rho\sigma}](x;x') = \text{diagram 1} + \text{diagram 2} + \text{diagram 3} \quad \text{COUNTER-TERMS}$$

The equation shows the 1-loop graviton self-energy on de Sitter space. The left-hand side is the self-energy function $-i[\mu\nu\Sigma^{\rho\sigma}](x;x')$. The right-hand side consists of three terms:

- Diagram 1:** A loop diagram with two external wavy lines (gravitons) attached to a circle at points labeled x and x' . The internal lines of the loop are labeled κ .
- Diagram 2:** A tadpole diagram with a single external wavy line attached to a circle at point x . The internal line of the loop is labeled κ^2 .
- Diagram 3:** A tadpole diagram with a single external wavy line attached to a circle at point x . The internal line of the loop is labeled κ^2 .

The text "COUNTER-TERMS" is written in blue to the right of the diagrams.